Assignment 2: Image Classification 2026
Due: Fri May 15, 2026 23:59
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Scenario

You are a member of a research team that develops image analysis methods for biomedical applications. Your team is currently designing new techniques to classify kidney conditions from CT scan images. Kidney disease is a significant global health concern, with conditions such as kidney stones, cysts, and tumours affecting millions of people worldwide. The creation of a rapid, automatic classification system could assist in earlier diagnosis and treatment, potentially improving patient outcomes and reducing the burden on healthcare systems.

You and a colleague have been tasked with designing and evaluating image classification algorithms for this system. In particular, you are to compare multiple algorithms and discuss them with reference to the state-of-the-art.

Task Overview

Note: This is a group assignment to be done in pairs (three students only if there is an odd number of students in the class).

In this assignment, you will be asked to demonstrate your ability to build classifiers for medical imaging data as part of a research study. You will be asked to build multiple classifiers for the same dataset and to compare them with each other as well as the state-of-the-art. The findings are to be summarised as a research paper. A key facet will also be to showcase your ability to discuss the advantages and limitations of biomedical image classification algorithms. You will be asked to produce three key things:

1. A research paper that describes your method and summarises your findings.
2. The code that you used to perform the classification.
3. A brief in-class demonstration or discussion of your findings.
4. Reflection on the assignment, a separate form to complete once the assignment is finished, and this can be submitted to a reflection portal.

5. **Report including Demonstration: 15%, Report: 20%; so, this assignment carry 35% of the final marks.**

Data

We will be using images from the study described in the following paper:

Islam, M.N., Hasan, M., Hossain, M.K., Alam, M.G.R., Uddin, M.Z., & Soylu, A. “Vision transformer and explainable transfer learning models for auto detection of kidney cyst, stone and tumor from CT-radiography.” *Scientific Reports* 12, 11440 (2022). [Available from the Library

Links to an external site.

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The dataset was collected from PACS (Picture Archiving and Communication System) from hospitals in Dhaka, Bangladesh, where patients were previously diagnosed with having a kidney tumour, cyst, normal findings, or kidney stones. Both coronal and axial cuts were selected from contrast and non-contrast studies using the whole abdomen and urogram protocol. The DICOM images were anonymised, converted to lossless JPG format, and verified by radiologists and medical technologists. The dataset citation is as follows:

Islam MN, Hasan M, Hossain MK, Alam MGR, Uddin MZ, Soylu A. Vision transformer and explainable transfer learning models for auto detection of kidney cyst, stone and tumor from CT-radiography. Sci Rep. 2022 Jul 6;12(1):11440. doi: 10.1038/s41598-022-15634-4. PMID: 35794172; PMCID: PMC9259587.

For this assignment, we will be using a collection of about 12,446 CT images across four classes. The breakdown of this full dataset is as follows:

- 5,077 Normal
- 3,709 Cyst
- 2,283 Tumour
- 1,377 Stone

You can use one of three versions of this dataset:

- **Full dataset of 12,446 images [ZIP, 1.6GB]**
- Download Full dataset of 12,446 images [ZIP, 1.6GB]
- **(NOTE: GIGABYTES)**
- **Partial dataset of 6,221 images [ZIP, 810MB]**
- Download Partial dataset of 6,221 images [ZIP, 810MB]
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- **Small dataset of 3,110 images [ZIP, 403MB]**
- Download Small dataset of 3,110 images [ZIP, 403MB]
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Note: The dataset does not come with a predefined train/test split. You will need to create your own split. You should clearly document the split and/or sampling strategy you use in your paper.

The structure of all three versions of the dataset is as follows:

- CT-KIDNEY-DATASET-Normal-Cyst-Tumor-Stone/
  Cyst/
  Cyst- (1).jpg
  Cyst- (2).jpg ...
  Normal/
  Normal- (1).jpg
  Normal- (2).jpg ...
  Stone/
  Stone- (1).jpg
  Stone- (2).jpg ...
  Tumor/
  Tumor- (1).jpg
  Tumor- (2).jpg ...

Tasks

Important: This dataset contains 4 classes. You are highly encouraged to do the 4-class classification problem (Normal vs Cyst vs Tumour vs Stone). You can also choose to perform the binary classification (Normal vs. Abnormal or Normal vs. one abnormal class). The class imbalance in the dataset is an important consideration that may be addressed and discussed in your paper.

1. **Implement several classifiers**
    1. Each team will train several classifiers with the given dataset.
    2. At a minimum, there should be 1 classifier per team member, and each classifier should be different (whether in the algorithm/technique used, or the feature set or both).
    3. Team members will benchmark the different classifiers with the given test dataset.
2. **Write a research paper**
    1. Each team will prepare a research paper describing their study. The paper will be formatted (fonts, margins, etc.) using the IEEE ISBI template
    2. Links to an external site.
    3. .
    4. The paper should be structured as follows. An approximate page guideline is provided.
        1. **Author Names and Introduction:** ~0.5-1 page
            1. A title and abstract are not needed. This should save you some space.
            2. The authors' names should be written in alphabetical order by surname.
            3. The introduction should identify and outline related work from the literature.
            4. The introduction should briefly compare and contrast the algorithms presented here in the context of the literature.
        2. **Technical Contributions:** 1-1.5 pages per team member
            1. This section should describe the classifiers used in the study and any processing done to the images before they were used for classification.
            2. Key steps should be justified.
            3. Each team member should write about at least one of the classifiers (the one they implemented). Use subsections and label each subsection with the name of the person implementing it ("Graph Method - Sandhya").
        3. **Evaluation:** 1.5-2 pages

        1. This section should describe the evaluation protocol that was used, and the outcomes of the experiments.
        2. The results describing the benchmark comparisons of the different classifiers should also be presented.
        3. An analysis or discussion of the results should also be presented.
    4. **Conclusion:** ~0.5 pages
        1. This section should summarise the key findings and identify possible directions for future work.
  5. The page limit will be a total of 6 pages for a 2-person team, and 7.5 pages for a 3-person team.
  6. An additional page (beyond the page limit) can be used <u>only</u> for **Acknowledgements** and **References (IEEE format)**.
3. **Discuss the results**
  1. During class, students will speak briefly about the classifiers they have implemented.
  2. Briefly describe each classifier and its input.
  3. As a group, discuss a comparison of the benchmark results. You may show one or more classifiers in action.
  4. The discussion should be about 7 minutes for a 2-person group, or 9 minutes for a 3-person group. You may use slides or showcase your code notebook.

Suggestions

- **Start the assignment early.** Depending on the amount of data you use and the algorithm you select, the processing can take quite a long time. If you attempt it too close to the due date, the code may not finish executing.
- Develop your code using a few of the images. Once you are sure the code is working, then train on the image dataset.
- Look at past papers in IEEE ISBI or other medical imaging conferences. These papers pack a lot of detail into a short piece of text.
- You do not have to use all the training data in any dataset and can instead sample a subset. You should state the data you used (whether a subset or a specific set provided), and discuss any impact this may have on your analysis.
- Your objective is not to get a perfect classification. Medical image classification is still an active research field with many different algorithms, with their own advantages and disadvantages.
- For stronger results and discussion, you should contrast your work with the research literature.

Submission Requirements

You have to submit a file containing your paper. The paper can be either in PDF or .docx formats, though PDFs are preferred.

As part of the submission, **each team member should upload the code for the classifier they implemented** as <u>a comment</u> on the submission. If using Google Colaboratory, you can download your code as a Python notebook (.ipynb). Marks will be assigned individually and this is a requirement to receive marks.

Assignment 2: Image Classification (Part A and Part B) Rubric

Criteria	Ratings					Points
Introduction and Related Work view longer description	**Excellent** The related work provides a sophisticated overview of the current state-of-the-art in biomedical image classification, highlighting the most prevalent methods related to the task and a brief outline of their capabilities or limitations. **4 to >3 pts**	**Good** The related work provides a good overview of the current state-of-the-art in biomedical image classification, highlighting the most prevalent methods that are related to the task in this paper. **3 to >2 pts**	**Fair** The related work provides a good overview of the current state-of-the-art in biomedical image classification, highlighting the most prevalent methods. **2 to >1 pts**	**Poor** There is a little introduction to the related work, or the related work is essentially a list of papers with limited analysis. **1 to >0 pts**	**Not done** No introduction section in the paper **0 pts**	/4 pts
Technical Contribution view longer description	**Excellent** The paper includes a description of the implementation of the classifier (code, functions, toolboxes) with a justification of the main steps, and also provides details on training.. Also, the paper describes the theory behind the classifier and justifies the theories used to construct it. **8 to >6 pts**	**Very Good** The paper includes a description of the implementation of the classifier (code, functions, toolboxes) with a justification of the main steps, and also provides details on training. **6 to >3 pts**	**Good** The paper includes a description of the implementation of the classifier (code, functions, toolboxes) with a justification of the main steps. **3 to >1.5 pts**	**Fair** The paper only includes details on how the classifiers were implemented (code, functions, and toolkits used). **1.5 to >0 pts**	**Poor** The paper does not include a technical description of the classifier, and how this was applied to the classification task. **0 pts**	/8 pts
Evaluation and Conclusion view longer description	**Excellent** The paper includes quantitative results on classifier performance using well-established metrics. The discussion section provides a sophisticated analysis of these results, including a comparison with related work, directions for future research, and other relevant content (as determined from the literature review). **6 to >4 pts**	**Very Good** The paper includes quantitative results on classifier performance using well-established metrics. The discussion section provides a robust analysis of these results, including a comparison with related work or directions for future research. **4 to >2 pts**	**Good** The paper includes quantitative results on classifier performance using well-established metrics. The discussion section provides a reasonable analysis of these results, focusing on where the classifier tends to work or fail. **2 to >1 pts**	**Poor** The paper includes a brief results section that shows examples of correct and incorrect classifications. The discussion section of the paper is sparse and does not much insight into the classifier performance. **1 to >0 pts**	**Incomplete** The paper does not include any meaningful results nor any written discussion of the classifier performance. **0 pts**	/6 pts
Live Discussion view longer description	**Excellent** The discussion provides a compelling description of the methods used, placing them in the context of the literature. There is a sophisticated discussion of the results, including the advantages and limitations of the algorithms used for this task. **15 to >12.5 pts**	**Very Good** The discussion explains the methods used, placing them in the context of the literature. There is a good overview of the advantages and limitations of the algorithms for the task. There is a clear demonstration/explanation of the classifiers. **12.5 to >10 pts**	**Good** The discussion explains the methods used, placing them in the context of the literature. There is a good overview of the advantages and limitations of the algorithms for the task. There may be a brief demonstration of one of the classifiers. **10 to >7.5 pts**	**Fair** The discussion briefly outlined the methods used in the paper, and gave an overview of the main results. **7.5 to >2 pts**	**Incomplete** The discussion was not conducted, or did not actually provide any useful information about the paper. **2 to >0 pts**	/15 pts
Expression and Clarity view longer description	**Excellent** The paper and discussion are both of a high-quality. While there may be minor issues, the majority of the explanations are articulate and clear. It reaches the quality of a scientific research paper. **2 to >1.5 pts**	**Very Good** The paper is clearly written. The audience could understand the discussion. There may be some areas of improvement, with regards to clarity of explanations. **1.5 to >1 pts**	**Good** The paper is generally clear and understandable. The structure is mostly logical, and the main ideas are communicated adequately. However, there are a few areas where explanations lack detail or clarity, and the analysis could be more coherent. With moderate revisions, the discussion could reach a higher level of quality **1 to >0.5 pts**	**Fair** The paper is difficult to understand. It is poorly structured and the analysis is vague and unclear. **0.5 to >0 pts**	**Poor** Not written in IEEE paper format **0 pts**	/2 pts

Pipeline:

Pipeline plan

Phase 0 — Shared infrastructure (build this together, Week 1, ~1–2 days)

This is the single most important piece for getting marks. Both of you evaluate against *the same* split and *the same* metric code. Anything asymmetric here and the comparison is worthless.

1. **Fixed train/val/test split** (shared artefact):
   - Stratified by class, seed-locked (`random_state=42`), saved as a CSV (`split.csv` with columns `filename, class, split`).
   - **70/15/15** — enough val for early stopping, enough test for stable CIs.

    - Both of you load from this CSV, never re-split.
    - **Known limitation** (must be in paper): no patient IDs → can't prevent adjacent-slice leakage. Acknowledge, don't try to "fix" it.
2. **Dataset choice** — start with **medium** (6,221 images). Meets Sandhya's CNN minimum, fits comfortably in Colab memory, classical ML runs in minutes on a laptop. If classical pipeline is fast enough, optionally also report on full at the end.
3. **Imbalanced by default** — keep the natural imbalance (matches clinical reality). Handle via class weights / weighted sampling, *not* downsampling. Report macro-F1 as primary metric (accuracy is misleading with imbalance).
4. **Shared evaluation harness** (one `evaluate.py` both of you import): accuracy, macro & weighted F1, per-class P/R/F1, confusion matrix, one-vs-rest ROC-AUC, **bootstrap 95% CIs** (1000 resamples of the test set), **McNemar's test** between the two final models' predictions.
5. **Shared preprocessing entry point**: a function that loads an image, resizes to a canonical size, returns numpy array. Each person can extend (Person A → grayscale + equalisation; Person B → ImageNet normalisation + augmentation), but they start from the same load.

Phase 1 — Person A: Classical ML

Loading medium dataset as grayscale, 256×256 (tunable).

- **Feature extraction** (concatenate all into a single vector):
    - First-order intensity stats (mean, std, skew, kurtosis, entropy)
    - **GLCM Haralick features** (contrast, correlation, energy, homogeneity, dissimilarity, ASM) at 4 angles × 2 distances = 48 features
    - **LBP histogram** (P=8, R=1, uniform patterns) — 10 bins
    - **HOG** (9 orientations, 8×8 cells) — downsample to keep dimension manageable
    - *Optionally Gabor bank — only if time permits; easy to add, not essential*
- **Dimensionality**: PCA to retain 95% variance, OR mutual-information feature selection.
- **Classifier candidates** (try all 3, report the best, discuss all):
    - SVM with RBF kernel (grid search `C`, `gamma`)
    - Random Forest
    - Gradient Boosting (XGBoost / LightGBM)
- **Tuning**: stratified 5-fold CV on train+val only. Touch test once at the end.
- **Interpretability story**: feature importance from RF / permutation importance. "The model decides based on texture features X, Y, Z" — this is the interpretability win over DL.

Runs locally in minutes. No Colab needed for Person A.

Phase 2 — Person B (you): Deep learning on Colab Pro

- **Dataset in Colab**: upload the `_medium` folder as a ZIP to Google Drive *once*, then at session start: mount Drive → copy ZIP to `/content/` → unzip to local SSD. Reading from Drive FUSE directly is 10–20× slower than local SSD.
- **Architecture**: **EfficientNet-B0** (ImageNet-pretrained). Picked over ResNet50 because: fewer params (5M vs 25M), better accuracy-per-FLOP, trains faster on Colab. Over Swin/ViT because you won't beat Islam's Swin number and you don't have the compute budget to fine-tune a transformer properly — the honest move is a competent CNN, well-trained.
- **Input**: 224×224 RGB (repeat grayscale channel), ImageNet mean/std normalisation.
- **Training protocol** (two-stage):

1. **Stage 1 — head only**: freeze backbone, train new classification head (global avg pool → dropout 0.3 → dense 4). Adam, LR 1e-3, ~5 epochs.
2. **Stage 2 — fine-tune**: unfreeze last ~2 blocks, LR 1e-5, up to 30 epochs with early stopping on val macro-F1 (patience 5).

- **Augmentation**: random flip (horizontal only — CT anatomy is not top-bottom symmetric), rotation ±15°, random zoom 0.1, random brightness/contrast ±0.1, **no vertical flip**. Rationale: matches Islam et al.'s augmentation for comparability.
- **Class weights**: inverse-frequency weights passed to CrossEntropyLoss (addresses imbalance without downsampling).
- **Optimizer**: AdamW, weight decay 1e-4.
- **Logging**: every run logs config + metrics to a JSON in Drive (reproducibility).
- **Checkpoints**: save best-val-macro-F1 checkpoint to Drive every epoch (Colab sessions can die).
- **Interpretability**: Grad-CAM on the last conv block, 2–3 examples per class, overlay on original image. Compare against Islam's Grad-CAM finding (VGG16 focuses well; ResNet/Inception dispersed).

Phase 3 — Shared: the analysis that wins marks

The headline accuracy isn't what gets you marks — these analyses are:

1. **Data efficiency curve**: train both models on 10% / 25% / 50% / 100% of the training data. Plot macro-F1 vs training size. Prediction: classical ML flattens early, DL keeps climbing. This is a *real* finding worth a paragraph.
2. **Per-class analysis**: Stone is the smallest class and hardest in Islam et al. — confirm or refute. Which model fails more on Stone?
3. **Confusion patterns**: what gets confused with what? (Cyst ↔ Tumor is the clinically important confusion.)
4. **Compute / wall-clock table**: training time, inference time, model size. Classical ML's selling point.
5. **Interpretability side-by-side**: feature importance (classical) vs Grad-CAM (DL), same 2–3 example images. Does the DL model look where a radiologist would?
6. **Statistical significance**: McNemar's test between the two final models. Bootstrap CIs on macro-F1.

Notes:

- Sandhya made clear that the understanding and justification of the choices, and the impact on the results are where the marks are allocated - not the quality of the classifier
    - All choices and decisions must be based on research

- If doing CNN, use the medium-sized dataset at least (can use large, but if computer too slow, then use medium)

- Each group member must make a different classifier (can be binary or multiclass classifier - base this on research)

- Do not submit more than 1 classifier per person. You can mention that you compared multiple, but only include the one that you finalise
- Can use small, medium, or large dataset depending on the computational power of your laptop

Anthony Thursday 23/4/26

- Research about training-test split to find best ratio
- Start with machine learning model